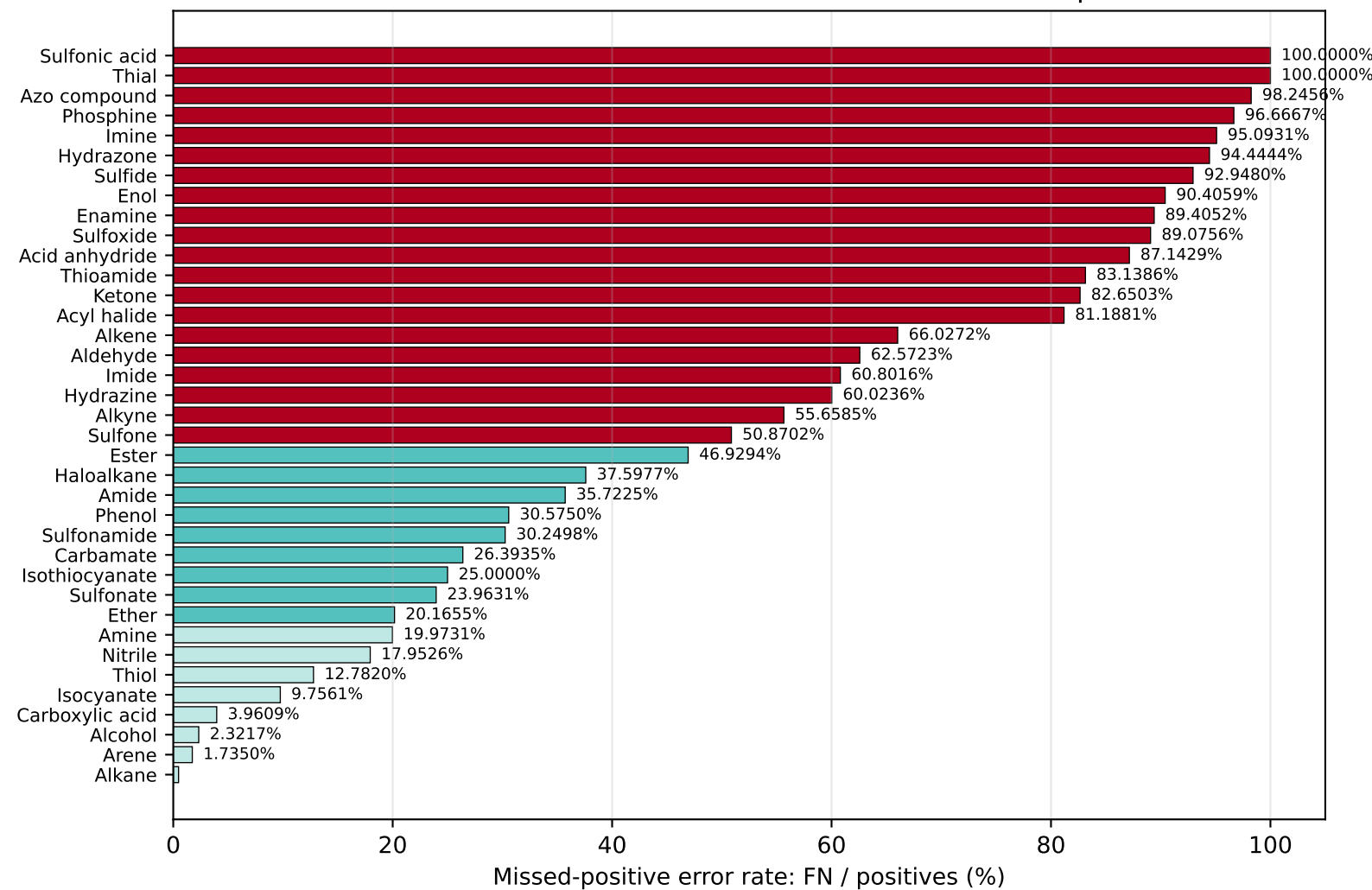
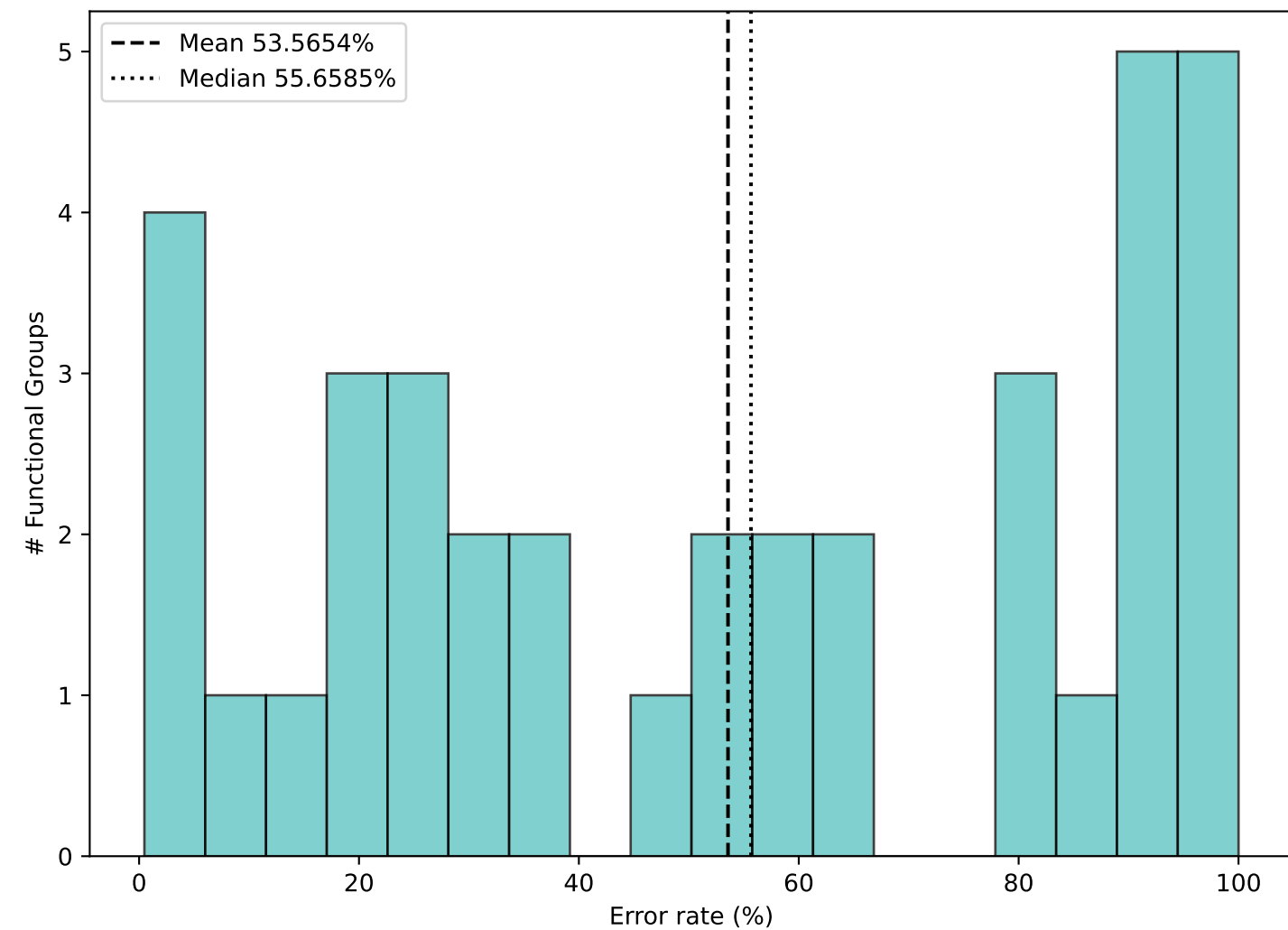


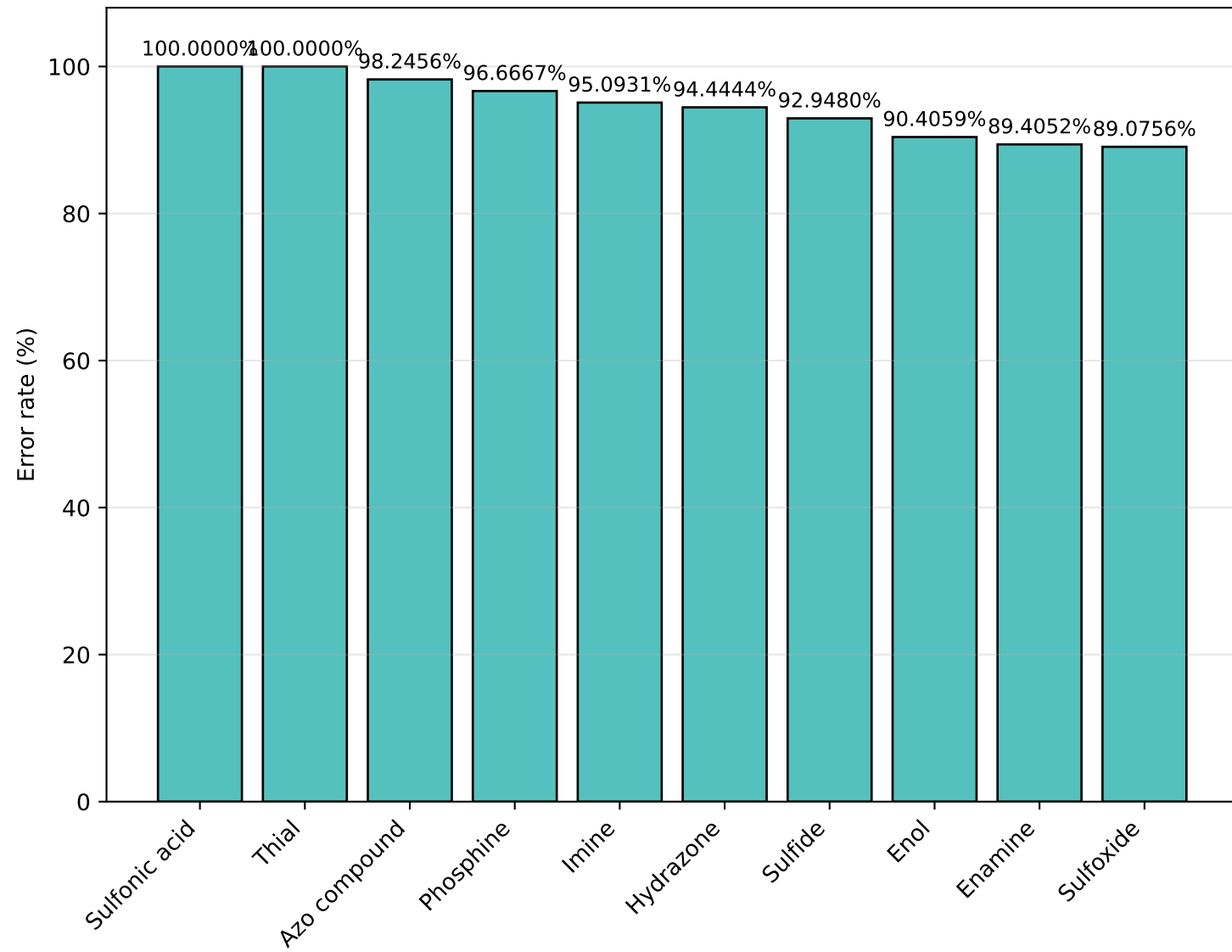
XGBoost IR Validation Set - All 37 Functional Groups



Error-Rate Distribution



Top 10 Problem Groups



XGB00ST IR VALIDATION SET ERROR ANALYSIS

Global metrics:

F1 micro:	0.8381
F1 macro:	0.5458
Precision micro:	0.8865
Recall micro:	0.7946
Hamming accuracy:	0.9603

Error-rate statistics:

Mean:	53.5654%
Median:	55.6585%
Std:	34.4699%
Max:	100.0000% (Sulfonic acid)
Min:	0.4834% (Alkane)

Problem classes:

Severe (>50%):	20 groups
High (20-50%):	9 groups
Moderate (5-20%):	4 groups
Low (<=5%):	4 groups